**Project Report**

**Title of the Project: AGRIBOT**

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# Abstract

Farming is growing smarter with the help of new technologies. This project, called *AGRI BOT*, introduces a smart farming robot designed to make planting seeds faster, easier, and more efficient. It combines modern tools with farming methods to bring better results with less effort.

AGRI BOT is a four-wheeled robot equipped with a robotic arm and a driller. It can be easily controlled through a mobile app using Bluetooth. The system is powered by an Arduino Mega 2560 microcontroller, which helps manage various parts like motors, sensors, batteries, and an LCD screen. The robotic arm carefully places seeds in the soil, while the driller makes small holes to plant them. Sensors guide the robot to move safely and follow the correct path.

The development of AGRI BOT follows a well-organized twelve-week plan, which includes identifying the concept, collecting components, building the robot, coding, and testing the system. The design is flexible, allowing future upgrades for tasks like watering, spraying, and removing weeds.

AGRI BOT makes planting more precise, saves time, and supports smart farming practices. It is helpful for both large farms and home gardens. This project shows how simple technology can make farming better and more enjoyable.

With user-friendly controls and the potential for adding more features, AGRI BOT is a great example of how technology can lead the way toward the future of farming.

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## Introduction to the project:

Agriculture is one of the most essential sectors contributing to food production and the economy. Over the years, various technological advancements have helped improve farming practices. Traditional farming methods are slowly being replaced by smarter techniques that save time, increase output, and require less manual effort. Farmers today are more open to adopting smart tools that simplify complex farming activities.

The use of robotics in farming is becoming more common around the world. Automatic systems for irrigation, harvesting, and even weed removal are now part of modern agriculture. This inspired the idea of building a smart robot that can assist in planting seeds. By integrating electronics and automation, the project aims to demonstrate how simple tools can play a big role in enhancing farming efficiency.

This project builds on the idea that technology can work alongside farmers to create a better and more productive environment. AGRI BOT is a step forward in this direction, combining creativity, engineering, and agriculture to make farming even more effective and enjoyable.

## Problem Definition:

Agriculture is a vital part of life, and with the right tools, it can become even more efficient and rewarding. Today, technology offers exciting possibilities to make farming tasks easier and more accurate. By introducing smart systems, farmers can enjoy better planting methods, balanced fertilizer application, and effective irrigation, all of which help improve crop growth and yield.

The AGRI BOT project was created with this idea in mind. It brings together modern tools and user-friendly design to support farmers in their daily work. This smart robot is designed to plant seeds with precision and can be controlled easily through a mobile application using Bluetooth. It uses an Arduino-based system, which is flexible and easy to upgrade.

By using AGRI BOT, farmers can complete planting tasks more quickly and consistently. It also encourages learning about new technologies in a simple and practical way. The project aims to create more interest in agriculture by showing how innovation and farming can go hand in hand to build a brighter future.

## Objectives of the proposed work:

The AGRI BOT project has been designed with a clear set of objectives to bring smart automation into seed planting. These include:

1. To develop a mobile-controlled robotic system for planting seeds accurately.
2. To simplify the farming process and reduce the need for manual labor.
3. To ensure proper seed spacing and depth for better crop growth.

4. To make the system affordable and easy to use for farmers.
5. To build a flexible design that can be expanded with additional features like spraying or watering.
6. To create a safe and reliable robotic model that can be operated in various farm environments.
7. To increase awareness and interest in agricultural automation among students and farmers.

By achieving these goals, the AGRI BOT will demonstrate the practical benefits of robotics in agriculture and how smart systems can make a real difference. The project also aims to inspire more innovation in this field and promote the use of technology to improve traditional farming practices.

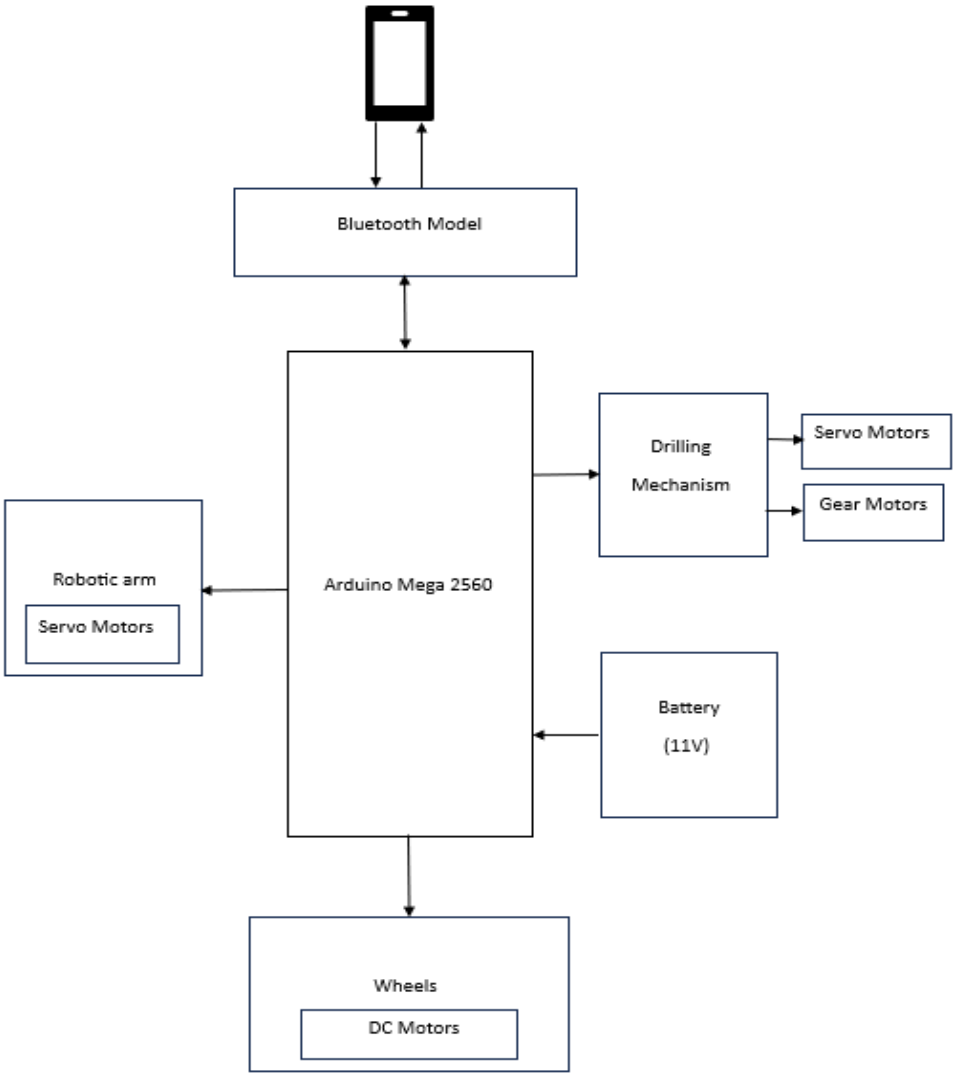
## Components:

- **Arduino Mega 2560 Microcontroller: For** central controller, managing all sensors, motors, and peripherals in the system.
- **HC-05 Bluetooth Module:** Enables wireless communication between the vehicle and an external device for control and monitoring.
- **LiPo Batteries:** Supplies lightweight and high capacity power to all components.
- **DC Motors:** Power the wheels to enable vehicle mobility and movement.
- **Servo Motors:** Provide precise angular control for robotics arm and driller movements.
- **Robotic Arm:** Handles tasks like seed placement or soil manipulation with precision.
- **Wheels:** Allow the vehicle to navigate across the planting area.
- **Wires**(Male to Male, Female to Female, Male to Female, Connecting wires): For connecting the components.
- **PCBs:** Provide a compact, organized platform for mounting and interconnecting electronic components.

- **Camera:** Identifies drilling spots, checks for obstacles, monitors drilling, guides the robotic arm for sapling placement, verifies alignment, and collects data for analysis.

- **Driller:** Used to create precise holes in the soil for planting seeds at specific depths ensuring proper placement for growth.

## Schematic of Agribot:



## Procedure:

The AGRI BOT project followed a structured approach, from component collection to final testing. The steps involved are as follows:

1. **Component Connection**: After gathering all the required parts such as the Arduino Mega 2560, motors, sensors, robotic arm, and HC-05 Bluetooth module, the components were connected carefully. Proper wiring ensured smooth communication among all parts.

2. **Code Execution Using Bluetooth App**: The mobile phone was connected to the HC-05 Bluetooth module. Using the Serial Bluetooth Terminal app, commands were sent as written in the code.

3. **Initial Robot Movement**: On receiving commands, the robot begins moving in a straight direction by default, or it moves as per the specified direction provided through the mobile app.

4. **Land Drilling**: As it moves, the drill is activated to dig the soil for planting.

5. **Seed Planting with Robotic Arm**: Once the drilling is done, the robot moves forward for a certain time and stops. The robotic arm then places the seedling into the drilled hole, using its grip to handle the seed.

6. **Reset and Repeat**: After planting, the arm returns to its original position, and the robot continues to move forward to repeat the process.

7. **Obstacle Detection with Camera**: A front-mounted camera helps detect obstacles in the robot's path. If any are spotted, the user can control or pause the robot using the mobile app.

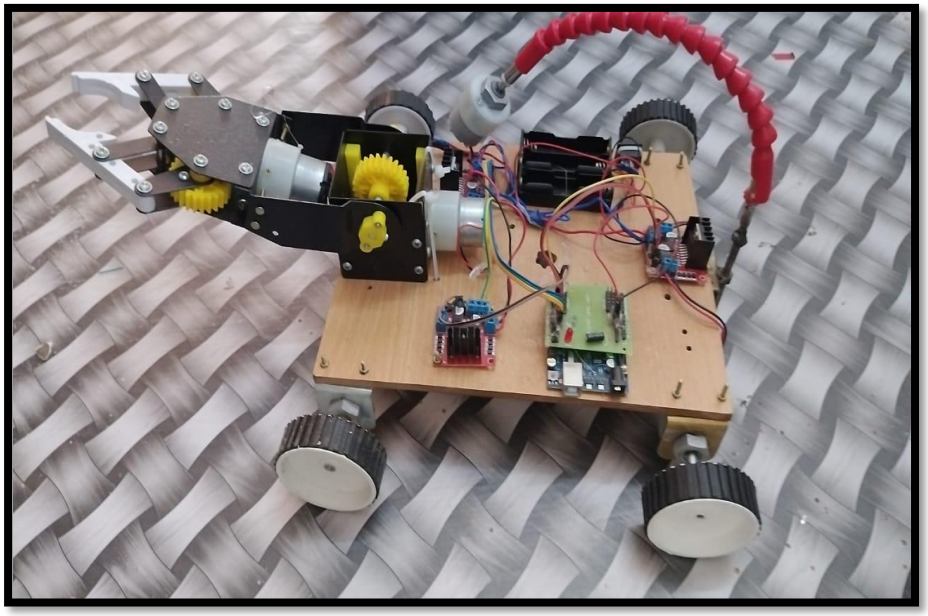
## Results and Discussion:

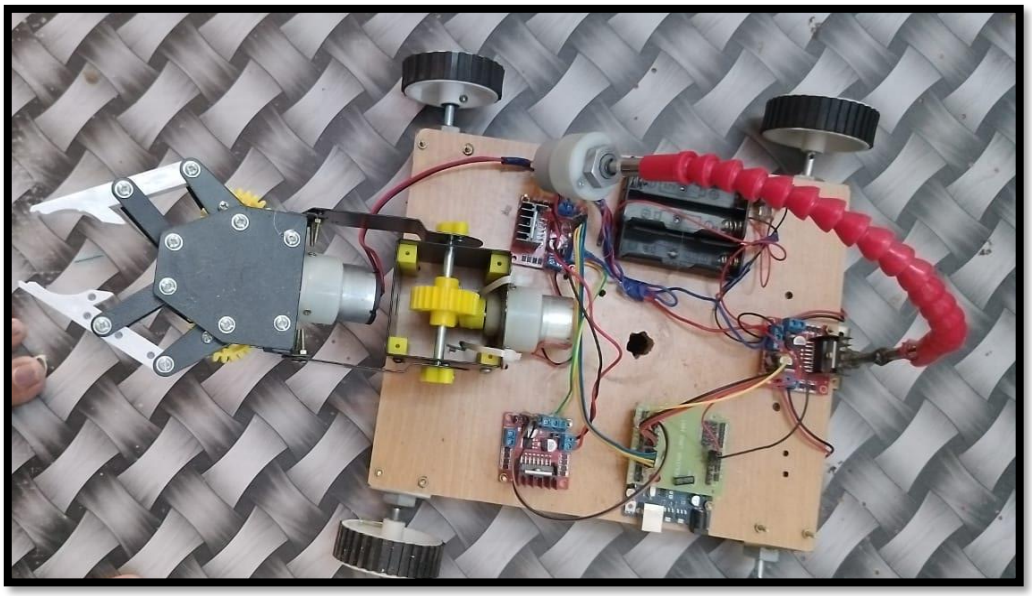
The AGRI BOT was tested across wet land, and performed each task successfully. The Bluetooth-controlled movement was smooth and responded accurately to commands sent from the Serial Bluetooth Terminal mobile app. The drilling mechanism operated with precision, creating a clear hole in the soil each time the robot stopped at the designated planting location.

The robotic arm showed good control while handling the sampling. It successfully placed the seedling into the drilled hole and returned to its original position. The robot continued its movement as designed, and the process repeated smoothly for multiple cycles. The obstacle-detection system using the front camera was effective, allowing the operator to manage the bot remotely when needed.

This automation allowed for uniform seed planting with minimal effort and consistent results. The overall performance indicates that the project is suitable for small-scale to medium-sized farms or for use in research and educational environments. It demonstrates how robotics can make agricultural tasks more structured and convenient, laying a foundation for further developments and results that the AGRI BOT can improve planting efficiency, save time, and encourage the use of smart farming practices.

## Images:





## Conclusion and Future Scope:

The AGRI BOT project has shown that integrating robotics into agriculture can bring significant improvements in productivity and convenience. By automating the seed planting process, it reduces the need for continuous manual work, helping farmers to save time while ensuring uniform planting. The project successfully combines mechanical design, electronics, and software into a single system that performs multiple agricultural tasks with precision.

The user-friendly control via mobile application and real-time Bluetooth communication makes it accessible and easy to operate. The implementation of features such as a robotic arm for planting and a camera for obstacle detection ensures a practical and efficient workflow. The project also emphasizes affordability and simplicity, making it a valuable tool not just for farmers but also for students and researchers interested in agricultural automation.

In the future, the AGRI BOT can be enhanced by adding capabilities such as automated irrigation, pest detection, and real-time soil analysis. Solar power integration could also improve energy efficiency. Additional improvements could include features for secure plant holding, weight optimization for easier field navigation, and redesigning the prototype into a full-sized functional machine with suitable wheels and robust materials for extended field use. These upgrades will expand the project's scalability and enhance its functionality, paving the way for a more advanced and accessible approach to modern farming.

## References:

Sahu, S. K., & Behera, S. (2021). *Solar Powered 4-Wheel Drive Autonomous Seed Sowing Robot for Rough Terrain*. This article introduces the proposed model and design of a solar-powered 4-wheel drive robot that plants seeds while cultivating the soil in multi-terrain surfaces.Research

Patil, S. S., & Patil, P. B. (2023). *Modern Agribot Seed Sowing with Enhanced Features*. This paper introduces a four-wheel-drive robot designed to autonomously sow seeds in ploughed agricultural land, alleviating the need for manual labour.Research

Kumar, R., et al. (2025). *Agriculture Seeding Machine*. International Research Journal of Modernization in Engineering Technology and Science. This project explores the development of an agriculture seeding robot controlled by a NodeMCU microcontroller and a motor driver, designed to automate the planting process and reduce labour.Research

Shamshiri, R. R., et al. (2021). *Design and Fabrication of an Agricultural Robot for Crop Seeding*. This research describes the development of a low-cost agricultural robot for crop seeding, detailing the prototype system's components and functionality.Research

## Code:

```
int m11=4;
int m12=5;
int m21=6;
int m22=7;

int r11=10;
int r12=11;
int r21=8;
int r22=9;

int dr1=2;
int dr2=3;
int md=1;
void setup() {
  Serial.begin(9600);
  pinMode(m11,OUTPUT);
  pinMode(m12,OUTPUT);
  pinMode(m21,OUTPUT);
  pinMode(m22,OUTPUT);

  pinMode(r11,OUTPUT);
  pinMode(r12,OUTPUT);
  pinMode(r21,OUTPUT);
  pinMode(r22,OUTPUT);
  pinMode(dr1,OUTPUT);
  pinMode(dr2,OUTPUT);
```

```
  delay(1000);
  digitalWrite(m11,0);
  digitalWrite(m12,0);
  digitalWrite(m21,0);
  digitalWrite(m22,0);

  digitalWrite(r11,0);
  digitalWrite(r12,0);
  digitalWrite(r21,0);
  digitalWrite(r22,0);
  digitalWrite(dr1,0);
  digitalWrite(dr2,0);

  delay(500);
}

void loop() {
if(md==0)
{
  Serial.println("clock");
  clk();
  delay(10000);
   Serial.println("Aclock");
  aclk();
   delay(10000);
    Serial.println("dstop");
   dstop();
```

```
delay(500);
 Serial.println("Front");
front();
delay(2000);
delay(500);
 Serial.println("stop");
stop();
delay(500);
 Serial.println("Pick");
pick();
delay(3000);
rstop();
delay(500);
 Serial.println("down");
down();
delay(5000);
delay(500);
 Serial.println("place");
place();
delay(3000);
rstop();
delay(500);
 Serial.println("rstop");
up();
delay(500);
 Serial.println("up");
delay(7000);
```

```
  rstop();
  delay(500);
   Serial.println("rstop");
  front();
  delay(500);
   Serial.println("front");
  delay(5000);
  delay(500);
   Serial.println("stop");
  stop();
}
 if(Serial.available())
 {
   int x=Serial.read();
   if(x=='1')
   {
     front();
   }
   if(x=='2')
   {
    back();
   }
   if(x=='3')
   {
     left();
   }
   if(x=='4')
```

```
    {
     right();
    }



    if(x=='5')
    {
      stop();
    }
if(x=='6')
    {
 place();
  delay(500);
  rstop();
    }
if(x=='7')
    {
   pick();
  delay(500);
  rstop();
    }
if(x=='8')
    {
  up();
  delay(500);
  rstop();
    }
```

```
  if(x=='9')
  {
 down();
delay(500);
rstop();
  }
   if(x=='A')
  {
 clk();
  }
   if(x=='B')
  {
 aclk();
  }
   if(x=='C')
  {
  dstop();
  }
   if(x=='D')
  {
  md=0;
  }
     if(x=='E')
  {
  md=1;
  }
}
```

```
}
void front()
{
digitalWrite(m11,1);
  digitalWrite(m12,0);
  digitalWrite(m21,1);
  digitalWrite(m22,0);
}
void back()
{
digitalWrite(m11,0);
  digitalWrite(m12,1);
  digitalWrite(m21,0);
  digitalWrite(m22,1);
}
void left()
{
digitalWrite(m11,1);
  digitalWrite(m12,0);
  digitalWrite(m21,0);
  digitalWrite(m22,1);
}
void right()
{
digitalWrite(m11,0);
  digitalWrite(m12,1);
  digitalWrite(m21,1);
```

```
  digitalWrite(m22,0);
}
void stop()
{
digitalWrite(m11,0);
  digitalWrite(m12,0);
  digitalWrite(m21,0);
  digitalWrite(m22,0);
}
void up()
{
  digitalWrite(r11,0);
  digitalWrite(r12,1);
}
void down()
{
  digitalWrite(r11,1);
  digitalWrite(r12,0);
}
void pick()
{
  digitalWrite(r21,0);
  digitalWrite(r22,1);
}
void place()
{
  digitalWrite(r21,1);
```

```
  digitalWrite(r22,0);
}
void rstop()
{
  digitalWrite(r21,0);
  digitalWrite(r22,0);
    digitalWrite(r11,0);
  digitalWrite(r12,0);
}
void clk()
{
  digitalWrite(dr1,1);
  digitalWrite(dr2,0);
}
void aclk()
{
  digitalWrite(dr1,0);
  digitalWrite(dr2,1);
}
void dstop()
{
  delay(1000);
   digitalWrite(dr1,0);
  digitalWrite(dr2,0);
}
```